

Single-Propeller Plane

Sahaj Patel

Georgia Institute of Technology

ME1670 Section A

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Project Description

As an aspiring aerospace engineer, I wanted to model and 3D print a single-propeller plane. I decided on this specific object because it reflects my passion for aviation, and it easily meets all the project criteria. The initial design sketches for the single-propeller plane are displayed in Figure 1.

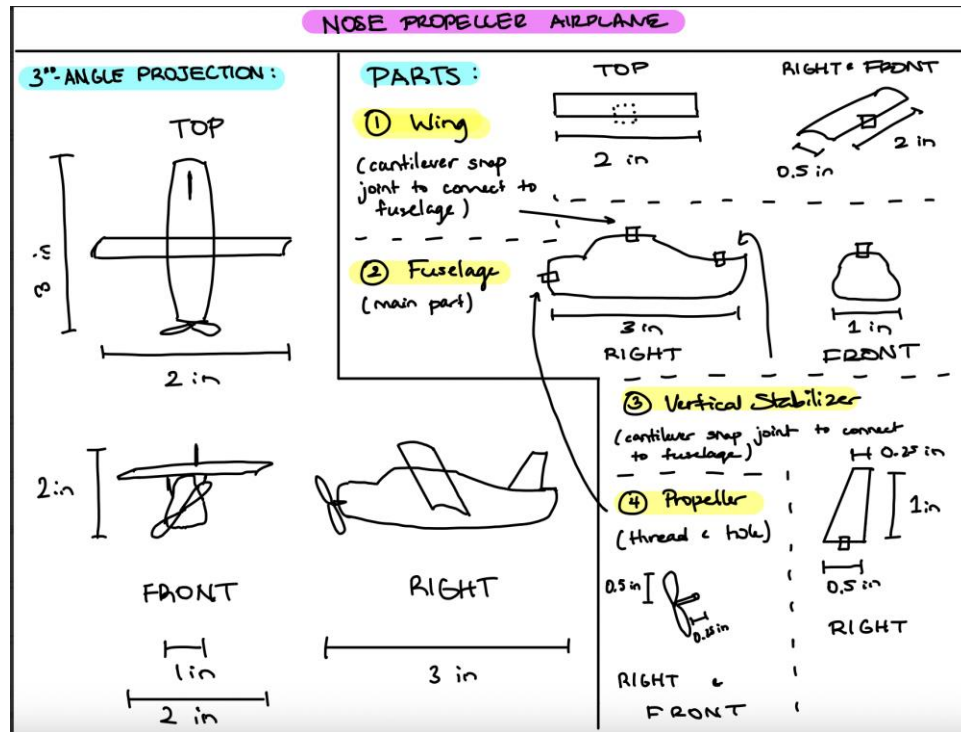


Figure 1: Initial Design Sketch

From the initial design sketches, there are two major changes that were made when developing the final design of the single-propeller plane. The first major change made was to incorporate the vertical stabilizer in unison with the fuselage. As seen in Figure 1, the vertical stabilizer was originally a separate part, but combining it with the fuselage allowed for a more streamlined structure and reduced the number of individual components, improving both stability and manufacturability. The second major change was the change of the propeller interlocking mechanism from a thread and hole to an axis-of-rotation snap joint. This was a very important change because the axis-of-rotation snap joint allows for the propeller to spin rather than being stationary and refrained from motion.

The final design of the single-propeller plane consists of three distinct parts: the fuselage, the wing, and the propeller. The fuselage is a solid body in which the wing and the propeller both interlock through snap joints. In addition, the fuselage contains indented windows and a vertical stabilizer on top of the main body. The wing, composed of a self-made airfoil, snaps into the fuselage via a cantilever snap joint located on the top face of the fuselage. The propeller, able to

spin freely, snaps onto an axis-of-rotation snap joint, which requires a squeeze so that the propeller can fit nicely.

Required CAD Features

Part 1 – Fuselage (shown in Figure 2)

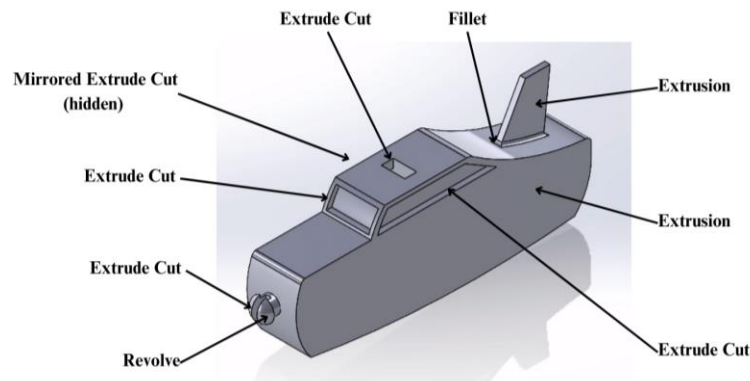


Figure 2 – Fuselage CAD Features

- Extrusion – Fuselage Body
- Extrusion – Vertical Stabilizer
- Extrude Cut – Cantilever Snap Joint Inlet
- Extrude Cut – Right Side Window Indentation
- Mirrored Extrude Cut – Left Side Window Indentation
- Extrude Cut – Front Window Indentation
- Revolve – Axis-of-Rotation Joint Outlet
- Extrude Cut – Axis-of-Rotation Join Pinch Slit
- Fillet – Vertical Stabilizer to Fuselage Smoothing

Part 2 – Wing (shown in Figure 3)

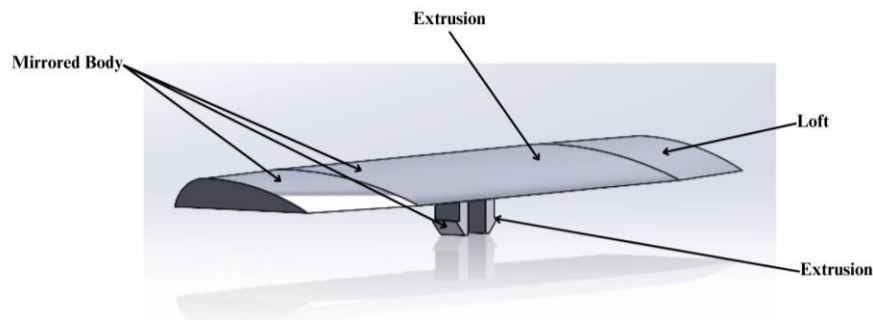


Figure 3 – Wing CAD Features

- Extrusion – Right Wing Center

- Extrusion – Right Cantilever Snap Joint Outlet
- Loft – Right Wing Outer
- Mirrored Body – Left Wing Whole

Part 3 – Propeller (shown in Figure 4)

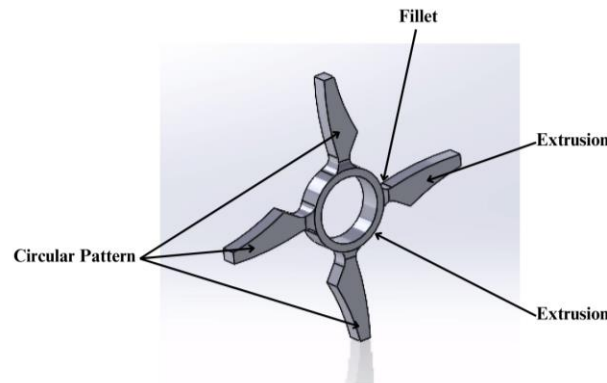


Figure 4 – Propeller CAD Features

- Extrusion – Propeller Blade
- Extrusion – Propeller Ring
- Circular Pattern – Three Additional Propeller Blades
- Fillet – Propeller Blade to Propeller Ring Smoothing

Design for Manufacturing Considerations

Interlocking Feature Decisions

One of the most important considerations when designing the single-propeller plane was to determine how each part would interlock with each other. During class lectures, various interlocking features were discussed, such as cantilever snap joints, axis-of-rotation snap joints, threads, gears, and hinges. In the final design, variations of snap joints were chosen for both interlocking features, as they best suited the intended function of each component. To connect the wing to the fuselage, a cantilever snap joint was used. This type of snap joint was selected because the purpose of the wing was to remain intact on top without moving. To connect the propeller to the fuselage, an axis-of-rotation snap joint was used. This type of snap joint was selected because the purpose of the propeller was to rotate/spin freely with ease.

Interlocking Feature Resources

The main resource when designing the aforementioned interlocking features was the lecture slides posted on Canvas. The lecture slides consisted of images where the application and construction of each interlocking feature were feasible to understand. In addition, searching for specific interlocking features on Google Images provided close-up insights into their dimensioning and construction, making it easier to implement them effectively.

Tolerance Values

For the interlocking features, the tolerance values used were 0.02-0.03 inches. Tolerance is crucial when designing CAD models because 3D printers are never perfect; prints could be slightly smaller or slightly larger than intended, which could change the overall functionality of the part/final design. Having a tolerance value of 0.02-0.03 inches reduces the risk of parts not fitting together properly while still maintaining a snug and secure connection. This balance ensures that interlocking features function as intended, accommodating minor variations in print dimensions without compromising the overall assembly.

Design Choices

To ease the manufacturing and printing process, certain design choices were made. The first design choice was the thickness. Having a sufficient minimum thickness is important for parts to not break or bend. The given minimum thickness requirement was 1 mm; however, the final design consisted of a minimum thickness of 3 mm to account for tolerance and to be safe. The second design choice was to consider the removal of the printing powder from internal cut-out parts. Removing powder from a larger hole is much easier than from a small one. In developing the cantilever snap joint for the wing, it was important to make the inlet large enough to facilitate effective powder removal. A third design choice was to incorporate fillets. Fillets allow for increased structural integrity and durability of a part by minimizing the risk of cracking and stress failure.

Potential Problem Areas

The interlocking features were the major problem areas of the single-propeller plane. Making sure each interlocking feature, in this case the two different snap joints, snapped perfectly in place was vital for the overall functionality of the final result. If the inlet for the snap joint was too big, the interlocking mechanism would fit too loosely; if the inlet for snap joint was too small, the interlocking mechanism would not fit at all.

To address these potential problem areas, a series of test prints were conducted. Using the Aero Maker Space and the MILL, test prints of the CAD files were done to see if the printed parts would interlock properly and work as intended. If any changes needed to be made dimension-wise, they were done accordingly.

Design Inspiration

The design was inspired by a Cessna plane, shown in Figure 5 and Figure 6. I wanted to create a simple aviation-related artifact, and a Cessna plane represents this ideation well.



Figure 5: Cessna Plane in Air



Figure 6: Cessna Plane on Ground

Using Figure 5 and Figure 6 as reference, I designed a similarly shaped fuselage. Cessna's have wings on top of the fuselage rather than through the fuselage, so I incorporated this design element into my design as well. Furthermore, most Cessnas have one propeller, so I made my design have one propeller as well. Overall, the final 3D single-propeller plane accurately depicts a Cessna plane.

Design Iterations

During the designing and testing processes, one significant iteration was made. Originally, the propeller would not fit into the axis-of-rotation joint on the fuselage due to the diameter of the propeller ring being smaller than the diameter of the extending joint. To fix this problem, the diameter of the propeller ring was increased so that it could overcome the extending joint and snap into place perfectly. The cantilever snap joint for the wing and the fuselage worked perfectly so no changes needed to be made there.

CAD Assemblies

CAD assemblies were utilized to evaluate the functionality of the design before proceeding with 3D printing. By assembling the components in CAD, potential misalignments, interferences, and tolerances could be assessed and adjusted as needed, ensuring a more efficient and accurate final print. One specific example was the propeller, where the assembly process helped verify its clearance and rotation, ensuring it would spin smoothly without obstruction.

Intermediate Models & Processes

Figure 7 shows the process from the intermediate models to the final model. At first, the propeller ring was too small to fit onto the axis-of-rotation joint extrusion on the fuselage, preventing the propeller to snap on and rotate. However, the final design consists of a bigger propeller ring so that it appropriately snaps onto the axis-of-rotation joint, allowing it to spin as intended.

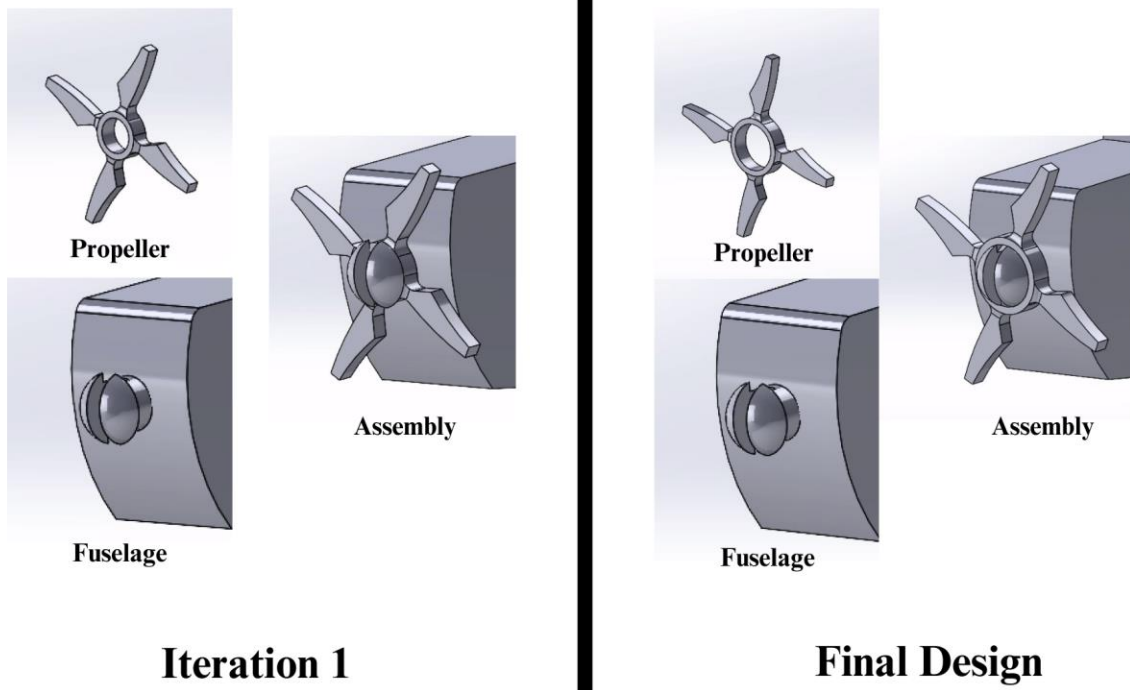


Figure 7: Intermediate CAD Models

In iteration 1 shown in Figure 7, the inner diameter of the ring was 0.16 inches, and the outer diameter of the ring was 0.21 inches. Noting that these dimensions were too small for the intended use, the final design, also shown in Figure 7, consists of an inner diameter of 0.21 inches and an outer diameter of 0.26 inches. These alterations fixed the previous problem and thus allowed the ring to spin freely about the axis-of-rotation joint.

Challenges Faced in Manufacturing

Overall, the manufacturing process went smoothly and as intended. The overall sizes and shapes of each individual part were depicted accurately. No scratches or abrasions were noticed following the manufacturing. Furthermore, the indented windows printed as planned, and the overall build was ultimately a true success.

During the assembly of the single-propeller plane, there were no significant difficulties. Every part fit perfectly – the wing perfectly snapped into the fuselage and the propeller rotated perfectly around the axis-of-rotation joint extruded from the fuselage. Initially, there was concern that the cantilevers might break due to the pressure required to snap the wing into the fuselage. However, the cantilevers fit perfectly into the wing without any breakage. In contrast, there was confidence that the propeller would seamlessly fit onto the axis-of-rotation joint, which proved true as it slid into place effortlessly. All of the dimensions and tolerances worked out favorably.

The only potential CAD change that would be made would be to decrease the inner diameter of the propeller ring by 0.01 inch to 0.20 inches. The propeller fits a little loosely on the axis-of-rotation joint and decreasing the inner diameter by a very small amount will allow the propeller to be not as loose. Other than this slight change, the rest of the CAD dimensions and output do not need to be altered.

To make the design process more efficient, feature patterns and higher tolerance values can be implemented. The use of feature patterns, such as linear patterns, circular patterns, and mirrors, can limit extra CAD work and redundancies, allowing for quicker designs. Having higher tolerance values reduces the risk for any interlocking part to not work as intended.

The CAD model had slightly different dimensions than initially planned due to small deviations introduced during manufacturing. However, the overall build remained fully functional because careful consideration of tolerance when designing the CAD was used. By accounting for acceptable variations in dimensions, it was ensured that the parts would still fit together and perform as intended. This highlights the importance of incorporating tolerance in engineering design to accommodate real-world manufacturing limitations while maintaining functionality. In addition, 3D printed parts will never be the exact same dimension as specified in the CAD, so considering tolerances is crucial for success.

In future designs, I will account for manufacturing tolerances to ensure that small deviations do not affect functionality. This includes specifying appropriate tolerance ranges, considering material behavior, and designing parts with proper clearances for assembly. By doing so, I can create manufacturable, reliable components that perform as intended despite minor dimensional variations.

Tolerance Analysis

Feature	Dimension in CAD	Actual Dimension	% Difference
Fuselage Length	2.78 in	2.829 in	1.747%
Fuselage Width	0.45 in	0.473 in	4.983%
Fuselage Height	1.00 in	1.117 in	11.053%
Wing Length	2.50 in	2.493 in	0.280%
Wing Width	0.70 in	0.688 in	1.729%
Wing Cantilever Extension Height	0.15 in	0.148 in	1.342%

Fuselage Axis-of-Rotation Joint Width	0.22 in	0.218 in	0.193%
Propeller Ring Inner Diameter	0.21 in	0.212 in	0.948%
Propeller Ring Outer Diameter	0.26 in	0.245 in	5.941%
Propeller Blade Height	0.30 in	0.276 in	8.333%

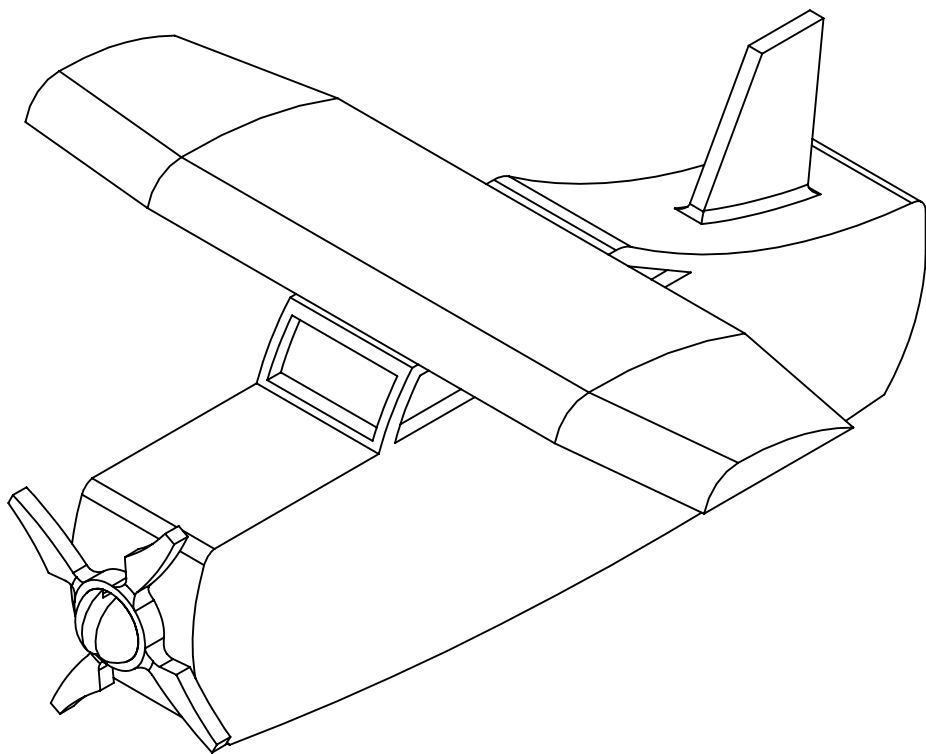
Table 1: Final Design's Dimensions

Table 1 portrays the differences between the original and final dimension values due to tolerance. As expected, there are many slight differences between the CAD and actual dimensions. Despite the dimensional changes, the overall functionality of the part works as intended. The fuselage height, though showing a significant difference, does not impact functionality since none of the other parts interact with it. The wing dimensions, while slightly different, still allow the parts to fit together as intended without affecting their ability to interlock. The axis-of-rotation joint width shows minimal variation, and this small difference does not affect its functionality. The propeller's inner and outer diameters, along with the blade height, do show some variation, but these differences do not prevent the parts from fitting or interacting properly. Overall, despite some dimensional deviations, the parts still assemble and function correctly without any significant issues in their fit or interlocking capabilities.

Conclusion

Overall, the process of designing, developing, iterating, printing, and assembling the single-propeller plane was an insightful experience that significantly enhanced my understanding of CAD modeling and 3D printing. Initially, the design began as a simple conceptual sketch with basic geometric shapes representing the fuselage, wings, and propeller. However, as the project progressed, several modifications were made to improve structural integrity, ease of assembly, and manufacturability. Additionally, interlocking features, such as snap joints, were refined to ensure a secure fit between components while accounting for tolerance in 3D printing.

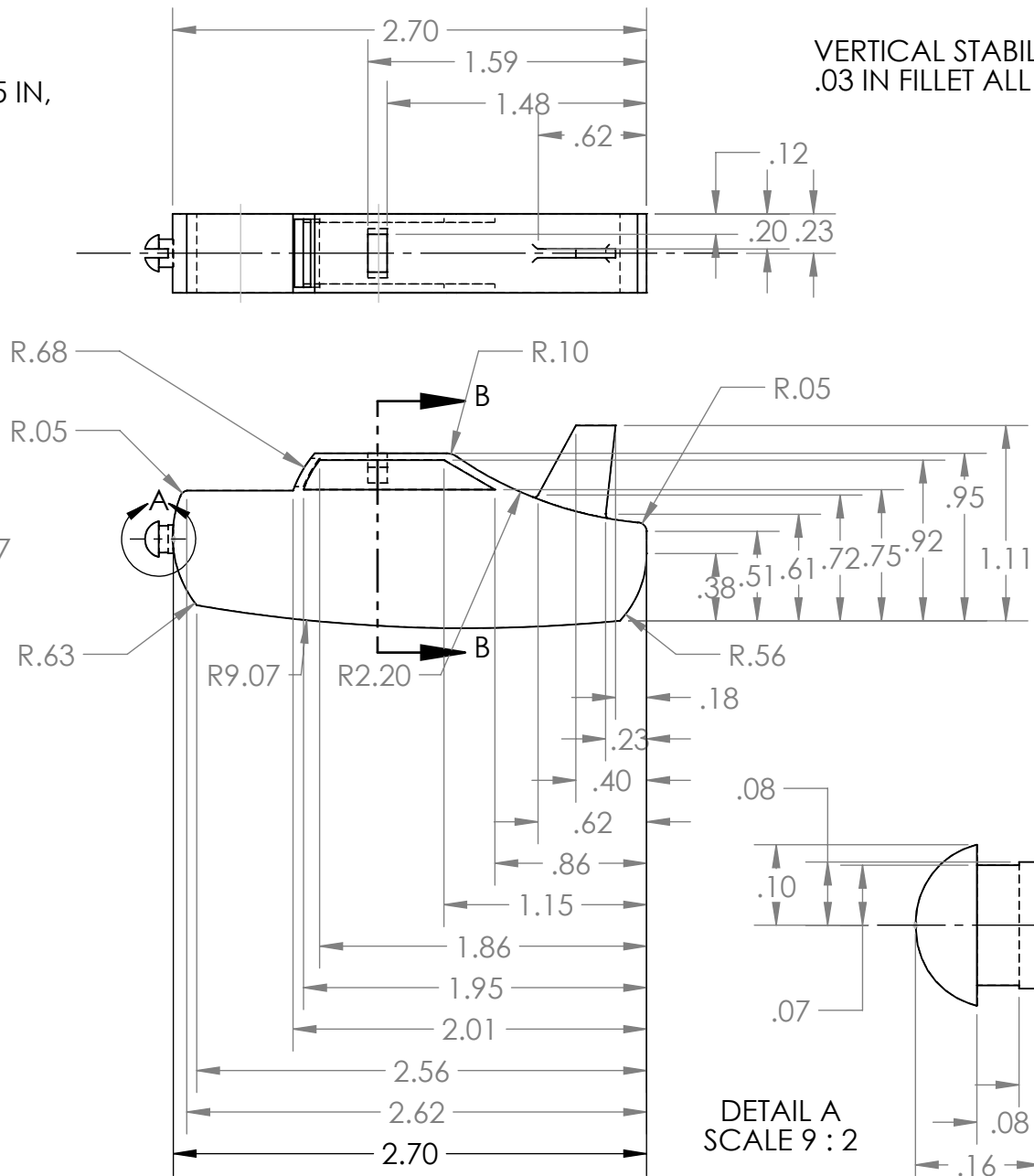
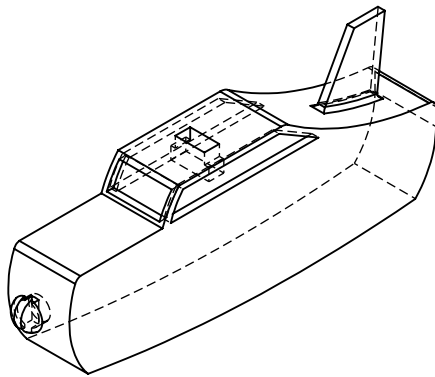
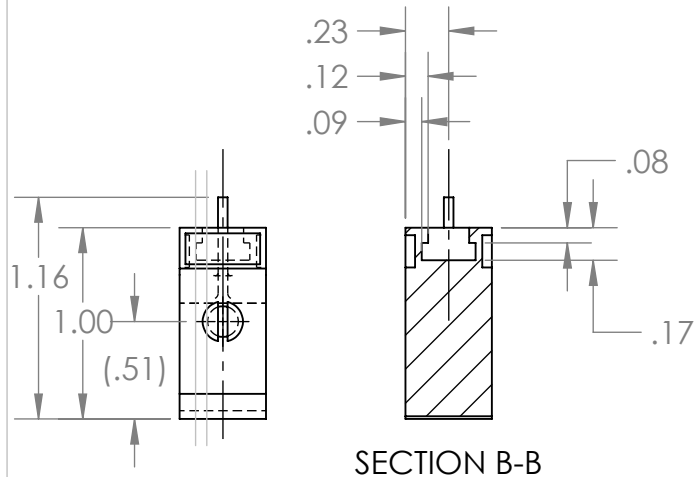
Through this iterative process, I learned the importance of designing with both functionality and manufacturability in mind. Small details, such as adjusting hole sizes to facilitate powder removal or refining tolerances for better interlocking, made a significant difference in the final assembly. The ability to troubleshoot and adapt throughout the design process proved to be the most valuable takeaway from this experience.



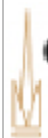
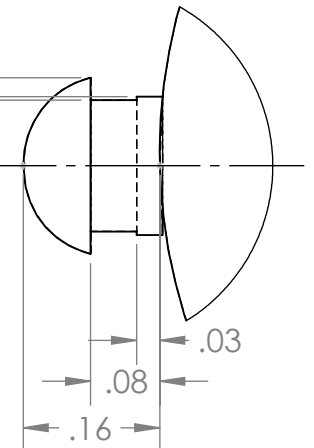
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UNIFORM WINDOW INDENTATIONS 5 IN,
UNLESS SPECIFIED OTHERWISE

VERTICAL STABILIZER
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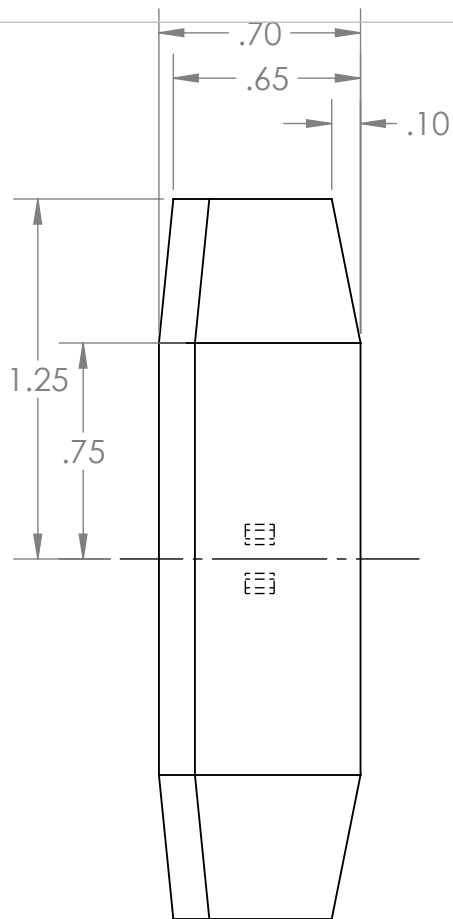


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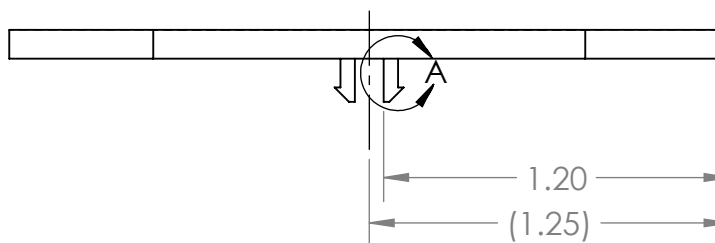
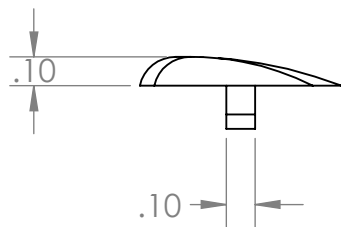
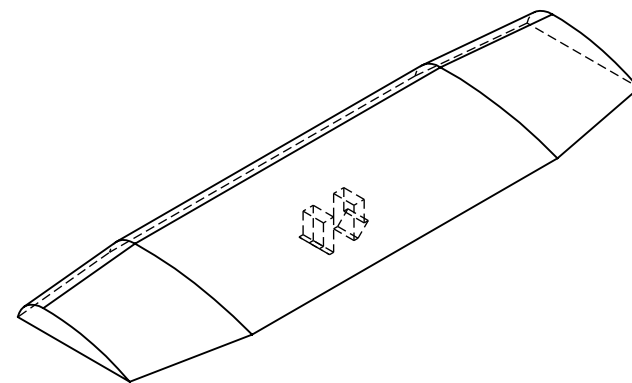
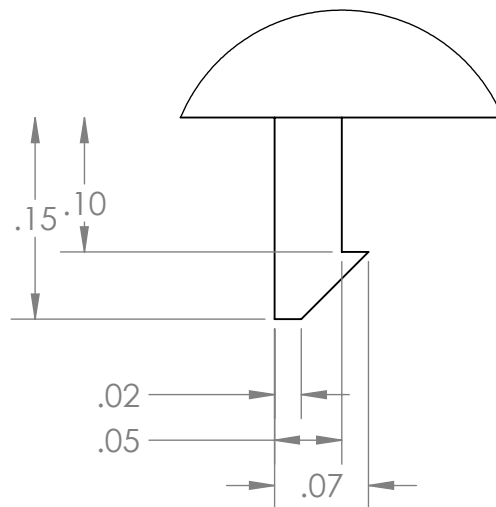
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DETAIL A
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TITLE
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ME 1670

SEMESTER
SPRING 2025

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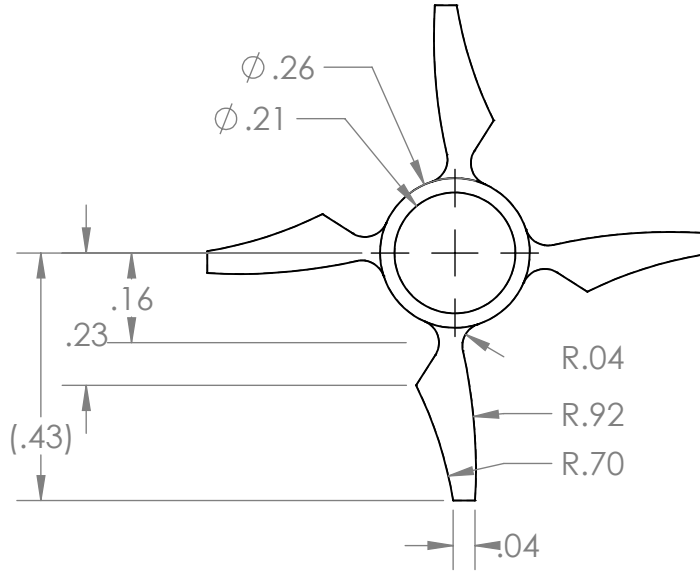
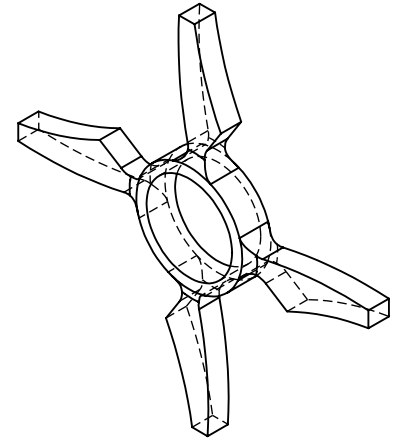
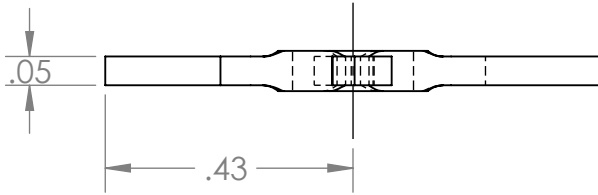
SECTION
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NAME
Sahaj Patel

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ANGLES ARE IN DEGREES
SCALE 3:2 SHEET 3 OF 4

SIZE
A



EACH PROPELLER BLADE HAS SYMMETRICAL DIMENSIONS



TITLE Individual_Project_Propeller	
COURSE ME 1670	SEMESTER SPRING 2025

DATE 4/18/2025	NAME SAHAJ PATEL
SECTION A	LAB
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